Supplementary Material: Literature Selection and Comparative Evidence

This supplement accompanies *Wheeled Mobile Robot Autonomy Under Degradation: A Survey of Methods and Evaluation*. It provides the selection and assessment details summarized in Section III, selected numerical comparisons discussed in the technical synthesis, and the tables of representative implementations referenced in Sections IV–VIII. Counts describe the selected literature and do not estimate research-topic prevalence across the field.

S1. Search Coverage and Selection

The initial search emphasized 2021–2026 publications across 17 journals and 102 journal–year combinations. It yielded 58,583 merged records, of which 43,086 had sufficient metadata for title/abstract screening. Language-model-assisted screening used preset topic allocations and stopped after 45 acceptances, leaving 43,041 metadata-complete records unassessed. Selection emphasized mechanisms relevant to compact wheeled platforms. The screening order and complete rejection history are unavailable. These missing records and the stopping rule limit the representativeness of the sample.

Additional coverage comprised 5 complementary publications, a 60-work thematic sample, 15 studies selected for the four integration questions developed in the survey, 20 IEEE publications concerning motion priors, recovery, shared-state diagnosis, and reproducibility, a further 48 targeted works, 36 recent IEEE publications, and 16 coverage additions. The 48-work stage covers degeneracy and map validity, constrained motion and terrain risk, multimodal navigation, communication-aware collaboration, and recovery or online learning. The recent IEEE stage selected six 2025–2026 IEEE journal or ICRA papers for each of the six themes after an exact-DOI check against the library. The selection table in Section III of the main paper summarizes these additions. The thematic sample contained 45 journal articles and 15 conference papers, with ten works under each of six themes: perception/localization, mapping/exploration, planning/control, multi-robot coordination, semantic navigation, and reliability/deployment. Searches combined technical, scenario, and mechanism terms with citation follow-up. Screening-based selection was completed by November 3, 2026, leaving coverage of the 2026 calendar year incomplete.

The final stage, on November 4, 2026, added 16 works from 2021–2024 that an internal review of an earlier draft named as missing. They comprise foundation-model and open-vocabulary navigation [1]–[5], uncertainty-aware terrain risk [6], [7], outlier-robust multi-robot SLAM [8], [9], DARPA Subterranean Challenge team systems [10]–[12], and datasets for degraded SLAM, off-road perception and dynamics, and embodied navigation [13]–[16]. They were not identified by a systematic search. The review also named VLMaps, whose 2025 journal extension is already in the corpus [17], NeBula, which exists only as a preprint, and further works that this stage did not acquire. Without these additions, 59 of the 228 technical publications (26%) report task outcomes under a named degradation condition, against 61 of 244 (25%) with them.

A. Journal and Conference Sources

The journal searches covered Automatica, Autonomous Robots, Control Engineering Practice, IEEE Robotics and Automation Letters, IEEE Transactions on Automatic Control, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Cybernetics, IEEE Transactions on Image Processing, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Intelligent Vehicles, IEEE Transactions on Robotics, Information Fusion, Journal of Field Robotics, Pattern Recognition, Robotics and Autonomous Systems, Science Robotics, and The International Journal of Robotics Research. The 2025–2026 searches covered 34 journal–year combinations. Completion of a search provider's enumeration does not establish omission-free retrieval of the publisher's catalog.

Formal CoRL volumes for 2024 and 2025 and RSS indexes for 2024–2026 were consulted. ICRA/IROS papers were assessed individually, without exhaustive enumeration of their proceedings. Complete enumeration of a formal CoRL 2026 volume was not established. Overlapping results from different search sources are not independent screening denominators.

S2. Eligibility and Publication Scope

Eligible publications include journal articles and main-conference papers concerning wheeled mobile robots or autonomy methods relevant to the navigation problems examined in the survey. Preprints and established conference-to-journal extensions are counted as one work, identified by title, authors, DOI or proceedings identifier, venue, and year. Publication-year statistics use formal volume years, whereas bibliography entries retain publishers' citation years, which can instead refer to online publication.

The selected literature contains 201 journal articles and 44 conference papers: 244 technical studies and one context review on learned navigation [25]. All corpus works are from 2021–2026, and two older studies held in the library are not cited. The 20 background references outside the corpus comprise representative estimators, related surveys, and foundational sources that attribute the definitions and equations of Section II: the kinematic classification of wheeled robots, the dynamic-window, timed-elastic-band, barrier-function, and

TABLE S1
SELECTED RESULTS AND WITHIN-STUDY COMPARISONS. TABLE AND FIGURE NUMBERS IN THE FIRST COLUMN REFER TO THE CITED PAPERS. EACH ROW RETAINS ITS OWN METRIC AND PROTOCOL.

Study / source	Protocol and measure	Reported result	Interpretation
LODESTAR [18] Tables III and V	SubT-MRS Long Corridor replay: absolute pose error (APE). Desktop computation averaged across benchmark recordings.	APE 1.386 m versus 3.113 m for FAST-LIO2; benchmark-average computation 28.69 versus 21.04 ms.	Selective anchoring and point selection improve this sequence at additional computation cost; no LiDAR-blackout or closed-loop test.
FIT-SLAM 2 [19] Tables 2 and 4	Exploration with simulator poses and common control; completion time, plus Fisher-information timing analysis.	Cave: 880 versus 1118 s for FAEL; Corridor: 1809 versus 1706 s. Planning: 20.8 ms without and 63.0 ms with Fisher information.	The ordering changes with the scene. Planner benchmarks and the SLAM/localizability ablation answer different questions.
Safe Horizon MPC [20] Table 2	100 Gaussian eight-pedestrian simulations; common solver/cost and matched joint-risk specification 0.05; duration mean (SD).	16.0 (1.9) s versus 18.4 (2.4) s for CC-MPC. Maximum estimated plan collision probabilities: 0.0092 versus 0.0019.	Faster traversal accompanies less conservative plan risk under a known motion distribution; these are not empirical mission collision rates.
LeLaN [21] Tables 2 and 3	Visible-target navigation: 150 trials, 28 objects, success within 0.2 m; separate 15-trial obstacle ablation.	Goal success 0.89 versus 0.67 for OWL-v2 + ZoeDepth. With obstacles: arrival 0.60 with versus 0.13 without collision training.	Target arrival and obstacle handling are separate evaluations; visible-target success does not establish open-world search or a collision-risk bound.
TAPM-IVP [22] Table 2	Scene 1 simulation; two robots, ten independent trials per method, common local planner/tracker.	Success 90% for each robot versus 40% and 30% in the two-robot ablation without partitioning and greedy subgoal selection.	The ablation supports the combined design, not either module alone. Rates are per robot; hardware demonstrations use two Ackermann vehicles.
GP-based recovery [23] Fig. 6 and Section V	Ten simulated runs per goal/world; same planner with and without recovery. Counts read from the plotted rates.	Dense 1 m doorways, goal 0: 8/10 versus 3/10 successes. Sparse 2 m doorways, goal 0: 10/10 for both.	Recovery benefit depends on the failure setting. The physical demonstrations do not provide a comparable success-rate estimate.
Home deployment [24] Table 3	Ten-home field deployment; three assistive-service types, including requested but unperformed interventions.	1130 successful, 177 failed, and 232 unperformed interventions out of 1539 requests. Reported success: 86.46% of 1307 attempts.	The success denominator excludes unperformed requests; availability and assistance conditions remain relevant to service-level interpretation.

Nav2 papers, a SLAM formulation, and a GNSS integrity review.

The selected literature includes 73 works involving road vehicles, other embodiments, handheld or mixed configurations, or no specified physical platform. These works inform mechanism comparisons, with platform differences considered when assessing transfer. The remaining 172 works also vary in drivetrain, sensing, and computational resources.

For the technical synthesis, publications are grouped by their principal contribution to localization/mapping, traversability/planning/control, semantic navigation, multi-robot coordination, or fault management. A publication may also inform other areas, although it appears only once in Fig. S1. These five functional groups differ from the six themes used to allocate the thematic sample.

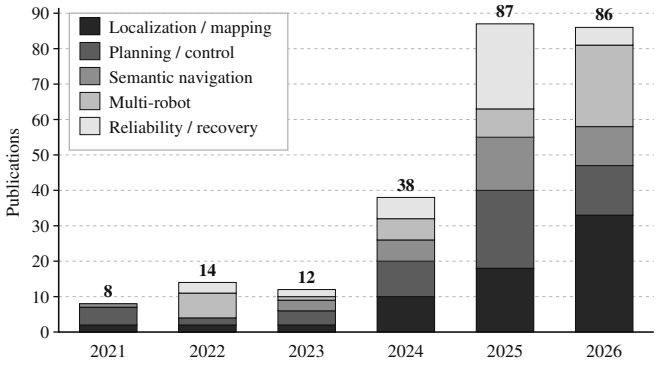


Fig. S1. Selected publications by year and principal navigation function. Each publication appears once. The distribution reflects purposive sampling and topic quotas.

S3. ASSESSMENT AND COMPARISON

Comparison attributes include the technical mechanism, platform, sensing, computation, experimental setting, baselines, measured outputs, and limitations. For 23 publications, assessment covered the full paper and numerical tables, whereas for 222 it was limited to selected method, evaluation, and conclusion passages or existing extracts. Of these 222 publications, 87 assessments were based on earlier extractions and 135 on targeted readings. These assessment levels do not provide uniform whole-paper verification across the selected literature.

The 48, 36, and 16 works of the last three stages received targeted readings with page-located excerpts matched to both the transcription and PDF text. Their evidence cards retain mechanism and protocol conditions. Numbers that the survey reports from these works were rechecked in the PDF text, without rendered-table audits. Four citation records are explicitly derived, including an early-access year correction and the replacement of an unparsable citation key. One further work has a documented official-title/PDF-title variant. These provenance exceptions remain visible separately from reading depth.

Experimental settings distinguish hardware control, online estimation, recorded robot data, dataset replay, simulation, and

numerical studies. Dataset and review publications are identified separately. Tables S2–S6 summarize mechanisms and inputs alongside experimental settings, evaluated quantities, and computation location. Interpretation accounts for external pose, remote computation, low speed, and staged scenes where applicable. Complete-pipeline timing and placement require information beyond processor identity.

Quantitative comparisons retain the reported metrics, baselines, and evaluation conditions. Effect sizes are not pooled across heterogeneous datasets, platforms, or trial units. The evaluation table of the survey counts attributes over the 244 technical publications, excluding the context review. It groups overlapping attributes by measurement level, test conditions, and risk or diagnostic assessment. A study may report component and task measures during a degradation test. Purposive selection, uneven assessment depth, and heterogeneous protocols limit the generality of comparative conclusions.

S4. Selected Results and Within-Study Comparisons

Table S1 retains the measures and experimental units of seven selected examples. Numerical results were checked against the cited tables or plots. Each comparison concerns the stated scene, baseline, and information available to the method. The rows therefore do not form a cross-study ranking. The examples include changes in performance ordering, added computation, combined-module ablations, and success denominators to clarify the scope of the conclusions drawn in the survey.

S5. Representative Implementations by Navigation Function

Tables S2–S6 list representative implementations for the five functional sections of the survey. Each row reports the mechanism and its inputs, the experimental setting with the evaluated quantities, and the platform with the computation placement stated by the authors. The rows were chosen to cover the response families discussed in each section, and they are not a ranking or a complete list of the selected works.

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TABLE S2
LOCALIZATION AND MAPPING: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
Uncertainty-aware localization [26]	Uncertainty-weighted prior-map matching with LIO and gated GNSS. *Inputs:* 3D LiDAR, IMU, GNSS	Acquired data *Pose error and covariance estimates*	wheeled (drive unspecified) *Unspecified*
LODESTAR [18]†	Selective historical anchoring and point selection under degeneracy. *Inputs:* 3D LiDAR, IMU	Replay *Absolute pose error and runtime*	mixed *Desktop evaluation*
ToF-Wheel [27]	Directional process-covariance inflation from geometric corrections. *Inputs:* time-of-flight camera and wheels; LiDAR replaces depth in replay	Acquired data, Replay *Trajectory error across covariance variants*	differential drive *Onboard*
DURAL [28]	Region- and degeneracy-dependent sensor fusion. *Inputs:* LiDAR, IMU, UWB, wheel encoders	Acquired data, Sim. *Localization error across fusion modes*	wheeled (drive unspecified) *Unspecified*
MM-LINS [29]†	Map hibernation, reinitialization, and pose-graph registration. *Inputs:* LiDAR/IMU from handheld and vehicle rigs	Acquired data, Replay *Trajectory error and map registration*	road vehicle *Unspecified*
SE2-LIO [30]	Near-planar pose with stochastic out-of-plane perturbations. *Inputs:* 3D LiDAR, IMU	Online estimate, Acquired data *Pose error; slope-model boundary*	wheeled (drive unspecified) *Onboard*
ProbPer-LiLo [31]†	Semantic-cluster persistence estimation and map filtering. *Inputs:* 3D LiDAR, semantic labels	Acquired data, Replay *Registration and map persistence*	mixed *Unspecified*
DMSA [32]†	Joint grid-distribution fitting over multiple scans. *Inputs:* 3D LiDAR, IMU	Acquired data, Replay *Pose error, runtime, sensor compatibility*	mixed *Desktop evaluation*
Switch-SLAM [33]†	Degeneracy-triggered introduction of visual constraints. *Inputs:* LiDAR, camera, IMU	Replay, Sim. *Trajectory error under modality failure*	mixed *Desktop evaluation*
M2UD [34]	Uneven-terrain localization and mapping benchmark. *Inputs:* LiDAR, RGB-D, IMU, GNSS; mixed reference systems	Dataset, Acquired data *Localization/mapping error and runtime*	wheeled (drive unspecified) *Unspecified*
DR-ICP [35]†	Directional feature selection and weighting. *Inputs:* LiDAR; capture IMU not integrated	Acquired data, Replay, Sim. *Tunnel pose error; pure-LiDAR scope*	mixed *Desktop evaluation*
Decoupled registration [36]†	Subspace decoupling and directional preconditioning. *Inputs:* LiDAR registration constraints	Acquired data, Replay, Numerical *Registration; missing-information boundary*	mixed *Desktop evaluation*
Semantic neural processes [37]†	Semantic neural-process elevation and uncertainty. *Inputs:* LiDAR, image semantics, poses	Acquired data *Elevation error and uncertainty; calibration needed*	road vehicle *Desktop evaluation*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

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TABLE S3
TRAVERSABILITY, PLANNING, AND CONTROL: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	**Mechanism and inputs**	**Evaluation setting** *Evaluated quantities*	**Platform** *Computation*
LRAE [38]	Unknown-region safety classification and route optimization. *Inputs:* 3D LiDAR	Hardware, Sim. *Exploration time and travel cost*	skid steer *Unspecified*
CADNet [39]	Heading-dependent accessible-range prediction for TEB planning. *Inputs:* LiDAR history	Hardware, Replay, Sim. *Traversability prediction and navigation*	wheeled (drive unspecified) *Onboard*
GA-Nav [40]	Semantic terrain segmentation combined with elevation costs. *Inputs:* RGB images, 3D LiDAR	Hardware, Replay *Terrain segmentation and navigation success*	wheeled (drive unspecified) *Onboard*
Anisotropic exploration [41]	Direction-dependent quasi-static terrain costs. *Inputs:* LiDAR terrain map, FAST-LIO pose	Hardware, Sim. *Exploration, travel cost, body roll*	differential drive *Onboard*
FIT-SLAM2 [19]	Keyframe-linked frontiers and information-aware adaptive roadmaps. *Inputs:* RGB-D/LiDAR (simulation); stereo depth (hardware)	Hardware, Sim. *Exploration time, pose error, and runtime*	skid steer *Unspecified*
Vehicle–cloud MPC [42]	Switching between onboard linear and cloud nonlinear MPC. *Inputs:* externally measured poses	Hardware, Sim., Theory *Tracking and collision outcomes*	Ackermann *Mixed placement*
CECBF [43]	CBF constraints with circulation and hold logic. *Inputs:* 2D LiDAR, external robot pose	Hardware *Obstacle avoidance and escape from equilibria*	differential drive *Local offboard*
Safe Horizon MPC [20]	Scenario constraints on joint collision risk with deceleration fallback. *Inputs:* robot/pedestrian poses and velocity estimates	Hardware, Sim. *Task duration and planned collision risk*	skid steer *Unspecified*
Spatiotemporal-Tubes [44]	Adaptive spherical tubes for prescribed-time reach-avoid. *Inputs:* robot state, online unsafe-set geometry	Hardware, Sim., Numerical *Reach-avoid and prescribed-time behavior*	omnidirectional *Unspecified*
Perceive With Confidence [45]†	Environment-level calibration of obstacle inflation for constrained planning. *Inputs:* RGB-D point clouds, VIO	Hardware, Sim. *Collision-free outcomes and calibration*	legged *Onboard*
Homotopy MPC [46]	Parallel optimization of homotopy alternatives. *Inputs:* externally tracked robot/pedestrian states	Hardware, Sim. *Dynamic avoidance; external pose*	wheeled (drive unspecified) *Unspecified*
Visual-geometric MPPI [47]	Expert-trained cost, speed, and novelty maps for MPPI. *Inputs:* RGB features, LiDAR geometry	Hardware, Acquired data *Interventions, speed, and terrain response*	wheeled (drive unspecified) *Unspecified*
SPARTA [48]	Angle-conditioned risk queried during planning. *Inputs:* prepared point cloud, LiDAR/IMU pose	Hardware, Sim. *Simulated tire risk; virtual risky hardware branch*	skid steer *Onboard*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

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TABLE S4
SEMANTIC AND LANGUAGE-GUIDED NAVIGATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
VLM-Social-Nav [49]	Cue-triggered VLM judgments used as social DWA costs. *Inputs:* RGB images, LiDAR	Hardware *Task success, collisions, and social behavior*	differential drive *Mixed placement*
Co-NavGPT [50]†	VLM frontier allocation with geometric goal execution. *Inputs:* RGB-D, shared map, robot states	Hardware, Sim. *HM3D search; separate hardware demos*	legged *Onboard pose; offboard perception/planning*
OpenIN [51]	Carrier–object memory with visibility-dependent updates. *Inputs:* RGB-D, language/image queries, SLAM pose	Hardware, Sim. *Success across sequential object queries*	wheeled (drive unspecified) *Unspecified*
VLFM [52]†	Image–language similarity projected into a frontier value map. *Inputs:* RGB-D, odometry	Hardware, Sim. *Goal success and path efficiency*	legged *Unspecified*
HOV-SG [53]†	Hierarchical retrieval over floor, room, and object graphs. *Inputs:* RGB-D, LiDAR-derived poses	Hardware, Sim., Replay *Object retrieval and navigation success*	legged *Unspecified*
Uni-NaVid [54]†	Multiscale video memory and discrete action prediction. *Inputs:* monocular RGB video, instruction	Hardware, Sim. *Goal success and path efficiency*	legged *Local offboard*
LeLaN [21]	Receding-horizon language-conditioned velocity prediction. *Inputs:* RGB image, text instruction	Hardware, Replay *Target arrival and action-generation time*	differential drive *Onboard*
VL-TGS [55]	VLM selection among generated trajectories with local DWA control. *Inputs:* LiDAR, RGB, velocity, GPS	Hardware, Acquired data *Outdoor traversal and model-response time*	wheeled (drive unspecified) *Mixed placement*
GSON [56]	Group-aware routing with predictive barrier-constrained control. *Inputs:* 2D LiDAR, RGB images	Hardware, Replay *Social disturbance, travel time, and jerk*	differential drive *Unspecified*
AVLMaps [17]	Multimodal spatial retrieval and goal APIs. *Inputs:* offline RGB-D/pose/audio map, instructions	Hardware, Sim. *Multimodal goals; prepared-map validity*	wheeled (drive unspecified) *Unspecified*
VLM-GroNav [57]	Interaction-grounded terrain costs and DWA. *Inputs:* RGB, LiDAR, proprioception, GPS, aerial images	Hardware *Outdoor task outcomes; GPS/cloud dependencies*	skid steer *Mixed placement*
SocialNav-SUB [58]	Social scene-understanding VQA benchmark. *Inputs:* recorded images, pedestrian annotations	Dataset *Human/rule agreement; no closed-loop control*	wheeled (drive unspecified) *Mixed placement*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

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TABLE S5
MULTI-ROBOT COORDINATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
ROAM [59]	One-hop consensus for semantic mapping and viewpoint planning. *Inputs:* RGB-D, LiDAR	Hardware, Sim. *Distributed mapping and exploration*	skid steer *Onboard*
Active graph exploration [60]†	Submodular loop-edge selection after coverage planning. *Inputs:* prior topo-metric graph	Sim. *Simulated coverage and graph objective*	none *Desktop evaluation*
Range integrity [61]†	Range-consistency localization-error reconstruction. *Inputs:* emulated ranges, reported poses	Online monitor (mixed reality), Sim. *Error reconstruction with emulated ranges*	aerial *Central ground station; emulated ranges*
Commerge [62]	Descriptor matching and scan selection before submap registration. *Inputs:* 3D LiDAR, LIO poses	Acquired data, Replay *Map-merging error and transmitted data*	wheeled (drive unspecified) *Desktop evaluation*
iMESA [63]†	Incremental local factor graphs with shared-variable consensus. *Inputs:* odometry and inter-robot factors	Replay, Numerical *Estimation error with limited communication*	mixed *Desktop evaluation*
Collaborative map repair [64]†	Residual-based fault isolation and correlation-aware map repair. *Inputs:* LiDAR, wheel odometry; GNSS initialization	Acquired data, Sim. *Pose/map error and fault isolation*	road vehicle *Unspecified*
TAPM-IVP [22]	Terrain-aware 3D Voronoi allocation and subgoal selection. *Inputs:* LiDAR, A-LOAM poses	Hardware, Sim. *Success, travel cost, and path duplication*	Ackermann *Onboard*
Failure-aware tracking [65]†	Failure-dependent tracking modes and target-state fusion. *Inputs:* target range/bearing and inter-robot messages	Hardware, Sim. *Target-state error during failures*	aerial *Unspecified*
Cross-rank [66]†	Intersection allocation from shared position histories. *Inputs:* LiDAR localization and low-bandwidth radio	Hardware, Sim. *Exploration under intermittent reception*	legged *Mixed placement*
CoVOR-SLAM [67]	Similarity-pose-graph fusion for monocular scale correction. *Inputs:* camera, inter-rover UWB ranges	Acquired data, Replay, Sim. *Pose error and monocular scale correction*	wheeled (drive unspecified) *Desktop evaluation*
Swarm-SLAM [68]	Sparse loop prioritization and decentralized inference. *Inputs:* local odometry, loop observations	Online estimate, Replay *Online collaborative estimation; team exchange*	skid steer *Onboard*
Planned synchronization [69]	Joint motion and observation scheduling. *Inputs:* relative marker observations, IMU	Hardware, Numerical *Formation and check-in cost; modeled non-idealities*	differential drive *Onboard*
Bandwidth-efficient SLAM [70]	Lossy keyframe selection and lossless coding. *Inputs:* visual/inertial keyframes	Online estimate, Replay *Bandwidth/accuracy tradeoff; central back end*	wheeled (drive unspecified) *Mixed placement*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

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TABLE S6
RELIABILITY AND RECOVERY: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
Visual NMPC recovery [71]	Tracking-loss stop and relocalization; restart after failure. *Inputs:* stereo vision, actuator feedback	Hardware *Tracking; stop/relocalization handling*	skid steer *Unspecified*
Set-based LPV diagnosis [72]†	Zonotopic detection and fault-class separation. *Inputs:* yaw rate, odometry, sideslip, steering angle	Acquired data *Detectability and isolability under injected faults*	road vehicle *Desktop evaluation*
Reliable MCL [73]	Joint pose/reliability estimation with global relocalization. *Inputs:* 2D LiDAR, odometry or inertial navigation	Acquired data, Replay, Sim. *Reliability estimates and relocalization*	wheeled (drive unspecified) *Unspecified*
Long-term home deployment [24]	Long-term service, docking, and remote human supervision. *Inputs:* LiDAR, RGB-D, docking and home sensors	Hardware *Service outcomes and interruptions*	wheeled (drive unspecified) *Mixed placement*
KSFD [74]	Kinematic residuals for sensor-fault diagnosis. *Inputs:* inertial channels and an inertial-aided position substitute	Acquired data, Numerical *Fault-isolation residual patterns*	Ackermann *Desktop evaluation*
Behavior-tree recovery [75]†	Behavior-tree supervision and failure-specific recovery. *Inputs:* terrain maps, contacts, planner status	Hardware, Sim. *Recovery on retained scenarios; human-rescue boundary*	mixed *Onboard*
Line-map integrity [76]	Prior-line-map localization with an error protection level. *Inputs:* monocular images, IMU, prior 3D lines	Acquired data, Replay, Sim. *Pose error and protection-level coverage*	wheeled (drive unspecified) *Desktop evaluation*
Event-triggered tracking [77]	Observer-based fault estimation and event-triggered tracking. *Inputs:* vehicle-posture outputs in the model	Hardware-in-the-loop, Numerical *Tracking error and update frequency*	Ackermann *Mixed placement*
Sentinel [78]	Action-consistency and VLM progress monitoring. *Inputs:* external RGB-D and mobile-base motion capture	Online estimate, Sim. *Failure detection and nominal alarms*	omnidirectional *Mixed placement*
RTEB [79]	Hybrid A* reinitialization and local smoothing. *Inputs:* RTK pose, voxel obstacle map	Hardware, Sim. *Planner recovery; TEB/NMPC comparisons*	wheeled (drive unspecified) *Onboard*
SCALE [80]	Affordance prediction and anti-novelty recovery. *Inputs:* RGB at deployment; offline robot data	Hardware *Visual recovery and intervention need*	skid steer *Onboard*
LiReN [81]	Online RL with resets, goals, and charging. *Inputs:* RGB policy; LiDAR/map, bumper for supervision	Hardware *Adaptation with operational resets and infrastructure*	differential drive *Mixed placement*

† Road-vehicle, non-wheeled, handheld, or mixed-platform evaluations.

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